

# Another look at Precision, Recall, and F1 score



COVID Test Classification Results

|               | Actually Has Covid | Doesn't Have Covid | Total     |
|---------------|--------------------|--------------------|-----------|
| Test Positive | 900                | 9,990              | 10,890    |
| Test Negative | 100                | 989,010            | 989,110   |
| Total         | 1,000              | 999,000            | 1,000,000 |

Source: Tom Chivers. Everything is Predictable, 2024.



"About 0.1% of people had Covid at the time [between the 1st and 2nd waves]. If you tested people at random, and your test correctly identified people who didn't have Covid **99% of the time**, and people who did have Covid **90% of the time**, **more than 90%** of your positives would be **false**."

..... The test correctly identifies 90% of the actual positive cases → **Recall**

: Out of all positive test results, only 8.3% are true → **Precision**

Is this a bad test? The F1 score is ~15%, which could indicate that it is. But in **low-prevalence settings** (aka class imbalance), high precision is almost impossible unless the test is nearly perfect.

Confusion Matrix

|          |              |
|----------|--------------|
| 900 (TP) | 9,990 (FP)   |
| 100 (FN) | 989,010 (TN) |

A confusion matrix helps make this more concrete:

- Precision =  $P(Y_{\text{COVID}+} | X_{\text{TEST}+})$
- Recall =  $P(X_{\text{TEST}+} | Y_{\text{COVID}+})$

**Precision** is the probability of having COVID given a positive test,  $P(Y_{\text{COVID}+} | X_{\text{TEST}+}) = 900 / 10,890 = 0.083$  (8.3%).

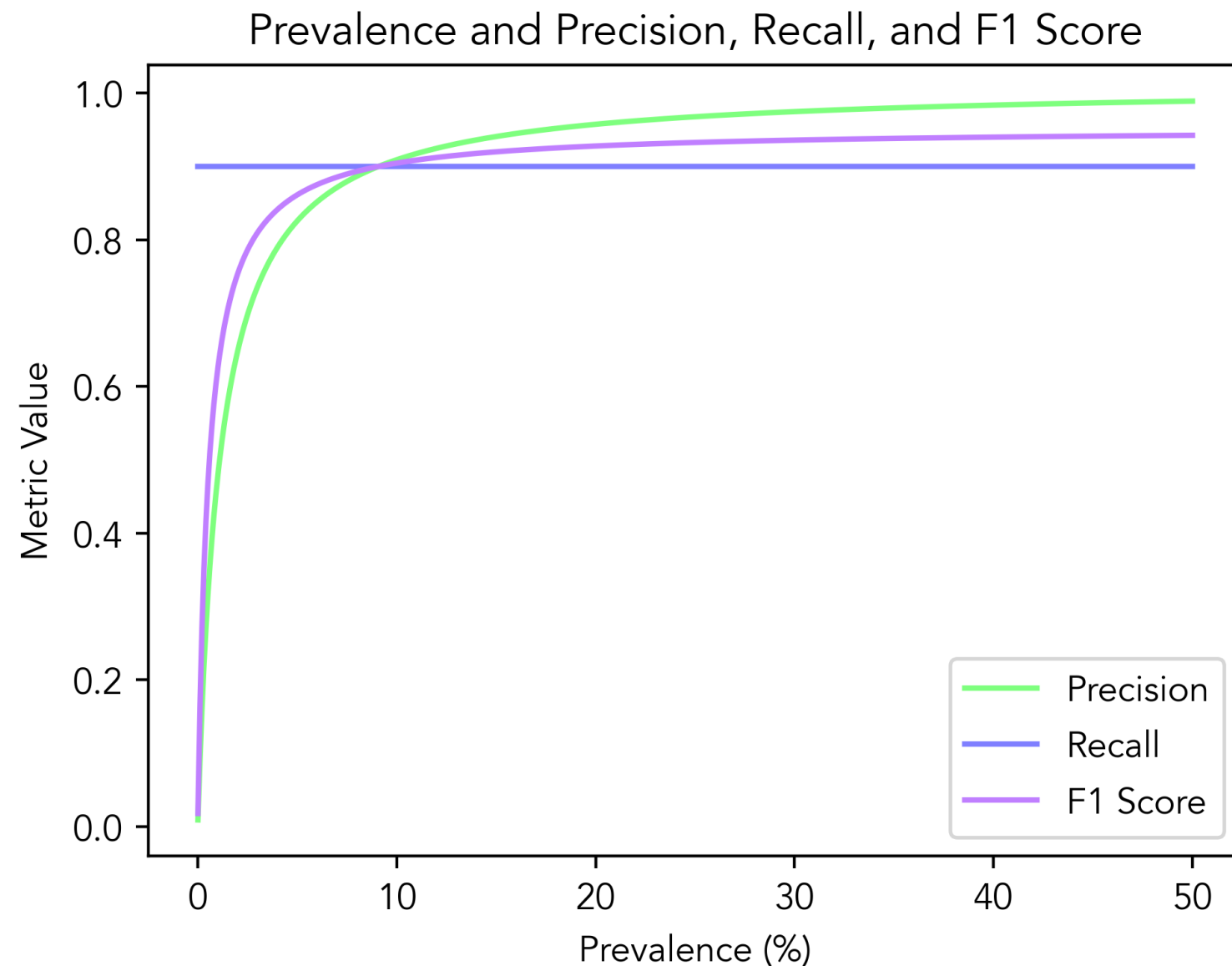
**Recall** is the class-conditional probability  $P(X_{\text{TEST}+} | Y_{\text{COVID}+}) = 900 / 1,000 = 0.9$  (90%).

**F1 Score** =  $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$   
=  $2 \times (0.083 \times 0.9) / (0.083 + 0.9) \approx 15\%$ .

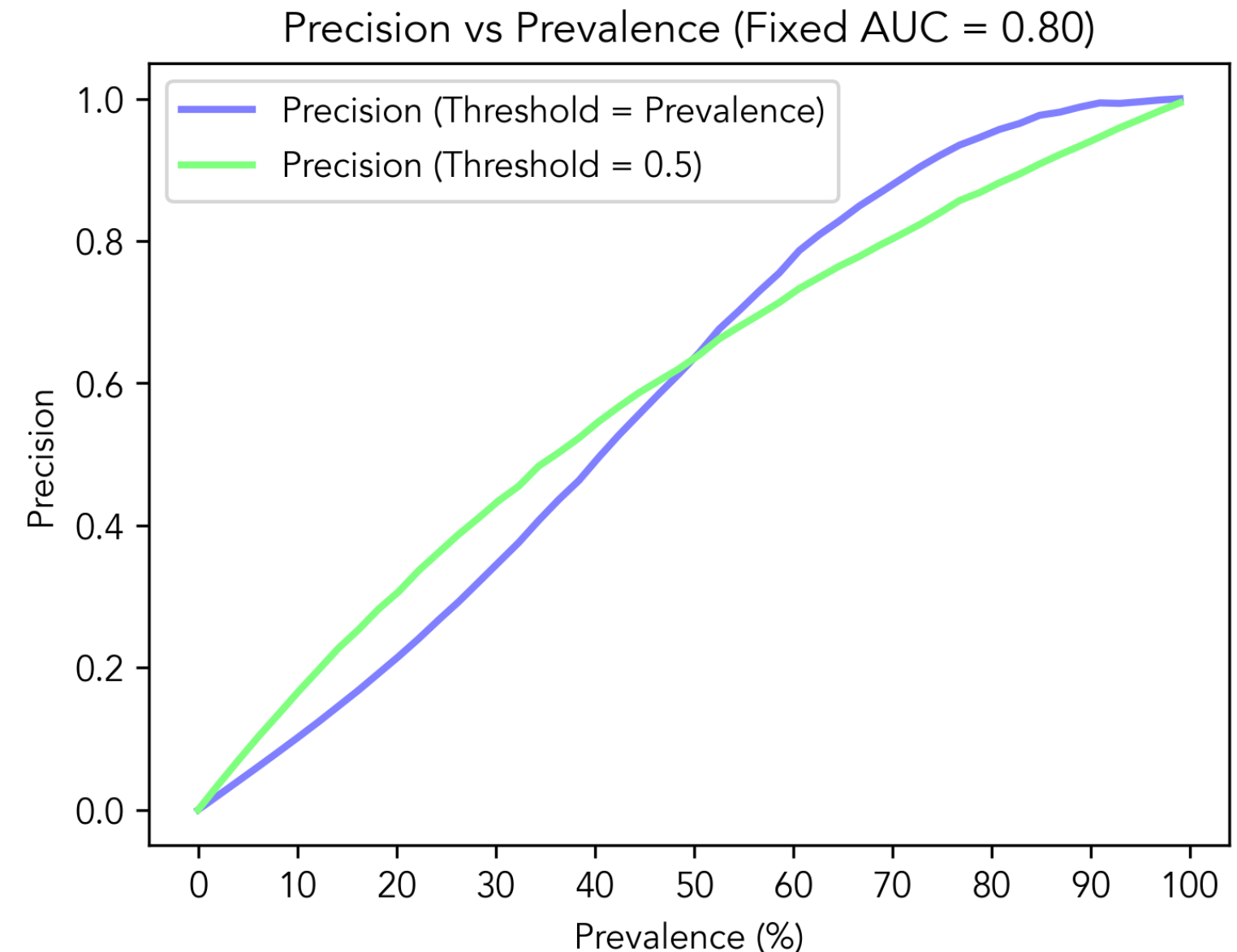
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Precision acts as a function of prevalence. When prevalence is very low, the denominator becomes dominated by **False Positives**, and Precision drops sharply.



The model's sensitivity (recall) and specificity ( $TNR = 1 - FPR$ ) are fixed, while prevalence varies.



The model's ROC AUC score is fixed, while prevalence varies. We use two types of thresholds to binarize predictions for precision scoring.

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We can show that prevalence is related to precision via the following formula:

$$\text{Precision} = \frac{\text{TPR} \times \text{Prevalence}}{\text{TPR} \times \text{Prevalence} + (1 - \text{TNR}) \times (1 - \text{Prevalence})}$$

Where TPR is True Positive Rate (Recall) and TNR is True Negative Rate. The denominator is the total positive rate, P(Test Positive).

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**Precision is also the probability that the outcome (or hypothesis) is true given evidence:**

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1 \mid Y = 1) \times P(Y = 1)}{P(X = 1)}$$

X = 1 is the positive test result, and Y = 1 is the presence of COVID. P(X = 1) is the total positive rate as shown below:

$$P(X = 1) = \frac{\text{TP} + \text{FP}}{N}$$

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